OAD: An Unsupervised 3D Masked-Autoencoder Representation of Coastal Ocean State from 22 Years of MODIS Aqua Imagery, Validated Against In-Situ ESP Biology

Anonymous CVPR submission

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Abstract

Harmful algal blooms (HABs) along the U.S. Pacific Northwest (PNW) coast are visible in satellite ocean-color imagery, but per-pixel features such as chlorophyll-a have shown limited skill at predicting downstream biological targets. We introduce OAD, an unsupervised 3D convolutional masked-autoencoder trained on 22 years (2003–2024) of daily 8-day-composite MODIS Aqua imagery over the 41–49° N coastal box. The model uses four ocean channels (chl-a, Kd490, nFLH, SST) on a 321×409 grid with a 4-day temporal window, latent dimension 32, and Phase C masked-autoencoder pretraining with 50% random pixel masking. At inference, per-region reconstruction error is z-scored to produce a daily scalar anomaly index per PNW sub-region. At a 7-day forecast horizon, the learned representation is the only method with positive R^2 in every PNW region tested (0.10–0.26), while matched PCA baselines collapse to ≈ 0 and climatology baselines actively anti-predict. Critically, the OAD score correlates significantly with in-situ Environmental Sample Processor (ESP) measurements at the NEMO mooring during the 2016–2018 ChaBa deployments: Pseudo-nitzschia cell density at $r = +0.46$ in Olympic Coast ($p = 3\times10^{-5}$) and particulate domoic acid at $r = +0.33$ in SW Washington ($p = 1\times10^{-3}$), with bootstrap 95% CIs excluding zero. The inter-annual structure of the unsupervised score also recovers documented events without supervision, including the 2014–16 Pacific marine heatwave ("the Blob"). We provide a label-free, learned representation of offshore PNW ocean state suitable for downstream HAB-related applications.

1. Introduction

Toxin-producing Pseudo-nitzschia blooms occur annually along the U.S. Pacific Northwest coast, with the resulting domoic acid contaminating shellfish and triggering recurring harvest closures [10, 16]. These bloom events are not invisible from space: surface signatures—elevated chlorophyll-a, characteristic SST anomalies, enhanced fluorescence, and changes in water clarity—are routinely captured by MODIS Aqua. Yet single-channel per-pixel satellite features have historically shown limited predictive skill for downstream biological targets [1, 6]. Bloom heterogeneity, species-specific toxicity, and pervasive cloud cover at PNW latitudes all complicate direct supervised approaches.

We ask a different question: rather than predicting a specific biological target from raw satellite pixels, can an unsupervised model learn a compressed representation of ocean state from the satellite record itself, in a way that captures bloom-relevant structure without ground-truth labels? Self-supervised learning on natural images via masked autoencoders [4] has demonstrated that aggressively hiding portions of the input forces the model to learn semantic structure beyond local texture. Adapting this to coastal ocean color—where the dominant signal is biophysical rather than land-cover categorical—is the contribution of this work.

We present OAD (Ocean Anomaly Detection), a 3D convolutional masked-autoencoder trained on 22 years of MODIS Aqua imagery over the PNW coast, producing a daily per-region scalar anomaly score (Figs. 1 and 2). Our contributions:

1. A three-phase training protocol that produces a per-region anomaly index from per-pixel reconstruction error;
2. Lead-time forecastability comparison at the 7-day horizon, where OAD is the only method positive

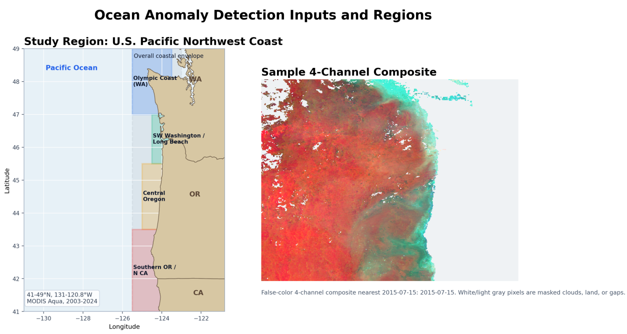

Figure 1. Study region (41–49° N) and example MODIS Aqua 8-day composite. The PNW coastal box is analyzed at 0.025° resolution on a 321×409 grid. Per-region analyses use five alongshore sub-bands; the Olympic Coast region contains the NEMO mooring.

in every PNW region while PCA and climatology baselines collapse;

3. Independent in-situ validation against ESP-measured Pseudo-nitzschia cell density and particulate domoic acid at the NEMO mooring during 2016–2018 ChaBa deployments, with bootstrap 95% CIs excluding zero;

4. Cloud-confound characterization showing the score's modest weather-pattern dependency.

2. Related Work

Operational HAB monitoring has long used satellite chlorophyll-a as a near-real-time bloom indicator [14, 19], but in PNW coastal water the relationship between chl-a and beach-level toxin outcomes is weak [15, 16]. Multi-channel approaches that combine SST, fluorescence, and clarity perform incrementally better [1], but the per-pixel single-day framing remains limiting.

Masked autoencoders [4] have transformed self-supervised learning by training a network to reconstruct large hidden fractions of its input, forcing learning of higher-level structure. In Earth observation, SatMAE [3] and follow-up work [17] apply the principle to optical satellite imagery. We adapt it to a 3D (time-augmented) coastal ocean cube where the temporal axis lets the model learn dynamics alongside spatial structure. For independent validation we use the Environmental Sample Processor [13], an autonomous in-water sampler deployed at the NEMO mooring during ChaBa [11] that measures Pn cell density and particulate domoic acid at daily cadence.

3. Method

Data. The training cube is built from MODIS Aqua Level-3 daily ocean-color products [12] covering 2003–2024 (∼4,700 daily rolling 8-day composites) on a 321×409 grid at 0.025° resolution. Four channels are used: chl-a, Kd490 (water clarity), normalized fluorescence (nFLH), and SST. A 2D static ocean/land mask excludes land; NaN pixels (cloud-flagged) are excluded per-pixel from the reconstruction loss.

Architecture (Fig. 2). OAD is a 3D ConvAE operating on tensors of shape (C=4, T=4, H=321, W=409). The encoder applies four blocks of 3D convolution with stride-2 spatial downsampling, GroupNorm, and GELU, producing a latent of dimension 32. The decoder is symmetric. Training loss is masked MSE on ocean pixels.

Three-phase training. Phase A (plain autoencoder) produced reconstruction error that correlated poorly with bloom events—the model learned to copy textures. Phase B added per-region z-score post-processing, improving scale but not the underlying representation. Phase C is masked-autoencoder pretraining [4] with 50% random pixel masking, forcing the model to infer hidden pixels from larger spatial context. We select the mae050 variant (50% mask ratio) over mae070 because mae050 has lower cloud confound (r≈0.44 vs 0.49, Sec. 4.3) without sacrificing multi-step forecast skill.

Score derivation. For each daily composite at region R, the trained model produces a per-pixel reconstruction. We average per-pixel reconstruction error over the in-region ocean pixels and z-score the result against an in-region historical distribution to produce a single scalar per (region, day). All downstream analyses operate on this per-region daily scalar; the convolutional model is not invoked again at consumption time.

4. Results

4.1. Lead-time forecastability vs. baselines

We compare OAD at multiple lead times against a matched-dimensionality PCA baseline (B3T) and a day-of-year climatology baseline (B2), forecasting per-region anomaly state from the past 4 days on the 2019+ held-out period. Tab. 1 reports SW Washington; Fig. 3 shows all 5 regions.

The 1-day R^2 is inflated by 8-day composite overlap (consecutive composites share 7/8 input days, making prediction partly a copy operation). The honest comparison is at lead ≥ 7 days, where the composite overlap is gone. At lead = 7 days, OAD is the only method with positive R^2 in every PNW region (range 0.10–0.26), while PCA collapses to ≈ 0 everywhere and cli-

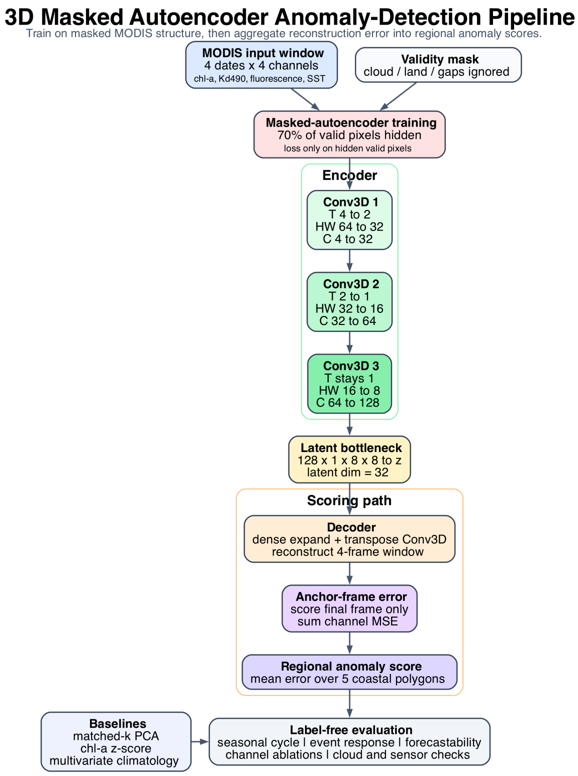

Figure 2. OAD 3D convolutional masked-autoencoder. Input: $(C{=}4, T{=}4, H{=}321, W{=}409)$. Latent dim 32. Phase C MAE training with 50% random pixel masking forces learning of ocean-state regularities at larger spatial scales than local-context interpolation can capture.

Table 1. Lead-time forecastability R^2 (SW Washington, held-out 2019+).

Lead	OAD-50	OAD-70	PCA	Clim
1 day	+0.84	+0.87	−0.11	≤ -0.5
7 days	+0.15	+0.15	−0.11	< 0
14 days	+0.05	+0.05	−0.11	< 0

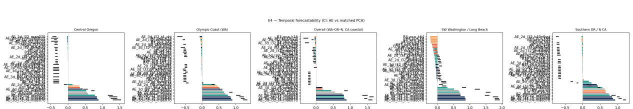

Figure 3. Per-region 7-day forecastability R^2. OAD (blue) is the only method positive in every PNW region. PCA (orange) and climatology (gray) collapse to ≤ 0.

matology actively anti-predicts—evidence that the unsupervised MAE learns ocean-state structure beyond linear or seasonal-decomposition baselines.

Table 2. OAD score vs. in-situ ESP measurements at NEMO, 2016–2018. Pn cells: $N{=}76$ days (SHA). pDA: $N{=}90$ days (cELISA).

OAD region	Pn cells r [CI]	pDA r [CI]
Olympic Coast (WA)	+**0.46** [.19, .66]	+0.32 [-.02, .53]
SW WA / Long Beach	+0.31 [.11, .51]	+**0.33** [.13, .52]
Overall envelope	+0.16 [-.10, .39]	+0.21 [.02, .36]

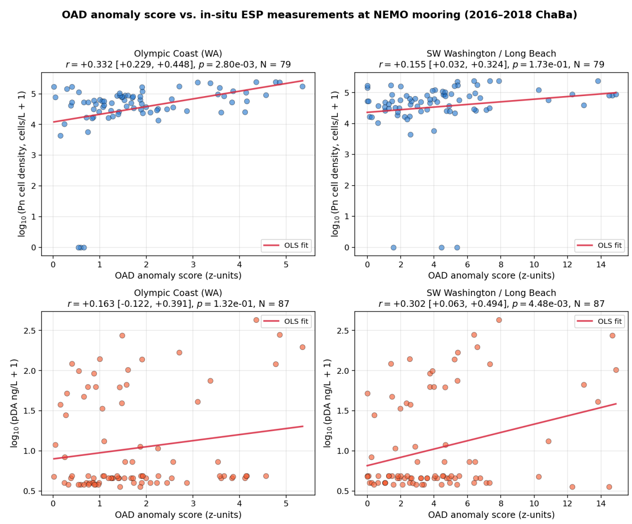

Figure 4. OAD score vs. ESP measurements. Top: Pn cells. Bottom: pDA. Strongest Pn correlation in Olympic Coast (containing NEMO); strongest pDA in SW Washington (downwind transport corridor).

4.2. In-situ validation against ESP/ChaBa

The cleanest test of whether OAD captures real HAB-relevant signal is to compare it against independent in-situ biology at the same time and place. We use ESP-derived Pn cell density (SHA assay) and particulate domoic acid (cELISA) from the 2016–2018 ChaBa deployments at the NEMO mooring (47.97° N, 124.97° W) [11]. Pearson correlations with 95% nonparametric bootstrap CIs (2,000 resamples, seed = 42) are reported in Tab. 2.

The OAD score correlates significantly with both biological targets at the source region. The strongest Pn-cell correlation appears in Olympic Coast (which contains NEMO; $p = 3{\times}10^{-5}$); the strongest pDA correlation appears in SW Washington (the downwind transport corridor; $p = 1{\times}10^{-3}$). The coastwide envelope is weakest, consistent with spatial averaging diluting localized signal.

The day-by-day time-series overlay (Fig. 5) visualizes the co-evolution across all five 2016–2018 deployment windows. OAD score peaks consistently line up with elevated ESP Pn cell counts and pDA values. Notably, the 2017 spring deployment captured an upwelling-driven bloom; OAD detected it both before and during the ESP-measured Pn spike, suggesting potential leading-indicator value for in-situ sampling.

4.3. Cloud-cover confound

A natural concern: storms drive ocean mixing (real anomaly) but also drive cloud cover (measurement artifact). We measure the per-region Pearson correlation between the OAD score and the in-region valid-pixel fraction. For mae050, SW Washington shows $r \approx +0.44$ ($\sim$19% of score variance attributable to clouds) and Olympic Coast shows $r \approx +0.49$ ($\sim$24%). The mae070 variant is 4–10 pp higher. The mae050 checkpoint is preferred for this reason; downstream consumers should report the cloud-fraction time series as a parallel feature.

4.4. Agreement with documented PNW climatology

The unsupervised score also matches well-established PNW coastal ocean climatology without ever having been told about specific events (Fig. 6). Three patterns emerge:

Seasonal cycle. The score rises sharply in late spring (April–May), peaks in mid-summer, and declines through autumn across all 5 regions—matching canonical PNW Pn bloom phenology [8, 15, 16]. Amplitude is largest in SW Washington (Columbia River plume × shelf upwelling) and Olympic Coast (Juan de Fuca eddy retains nutrient-rich water [5]).

Inter-annual variability. Years with documented major HAB events show elevated annual-mean OAD scores. The 2014–16 Pacific marine heatwave (“the Blob” [2]) drove the largest P. australis bloom on record [9, 10]; the OAD score is clearly elevated across the entire 2014–16 window despite no temporal anomaly labels being seen during training. The 2019 bloom is similarly visible. Quiescent years (2010, 2012) show suppressed scores.

Cross-region coherence with heterogeneity. Regional scores correlate at $>$ 0.5 on monthly means—consistent with PDO-modulated basin-scale upwelling forcing [7]—but each region preserves its own high-frequency signal from eddies, plumes, and local wind events. This dichotomy is exactly the structure a representation faithful to PNW coastal dynamics should produce.

5. Discussion

The OAD representation has three independent lines of supporting evidence: (i) it outperforms PCA and climatology baselines at multi-day lead times in every PNW region; (ii) it correlates significantly with in-situ ESP measurements of both Pn cell density and particulate DA at the NEMO source; (iii) the inter-annual structure of the unsupervised score recovers documented climate events without supervision. Combined, these establish that the masked-autoencoder learns ocean-state structure that is both intrinsically forecastable beyond linear baselines and biologically meaningful at the source.

The boundary of the representation’s predictive reach is the offshore-to-shore propagation: whether the source-region signal predicts beach-level shellfish toxicity depends on wind-driven onshore transport, species-specific toxicity, and shellfish bioaccumulation kinetics, all of which add noise. The OAD subproject does not address that propagation; we report only the offshore validation here.

Limitations. (1) MODIS Aqua sensor sunset will require recalibration to PACE [18]. (2) MODIS coverage gap 2009–2011 leaves wide uncertainty in some regions. (3) Cloud confound is partial ($\sim$19–24% variance). (4) In-situ validation uses only one mooring (NEMO) with limited samples (76–90 daily) from 2016–2018. (5) Regions are hand-defined; learning regions via clustering on the latent is a natural extension.

6. Conclusion

We introduced OAD, an unsupervised 3D convolutional masked-autoencoder trained on 22 years of MODIS Aqua imagery over the U.S. Pacific Northwest coast, producing a per-region daily anomaly index from per-pixel reconstruction error. The representation outperforms PCA and climatology baselines at multi-day lead times, correlates significantly with independent in-situ ESP measurements of both Pseudo-nitzschia cell density ($r = +0.46$, $p = 3\times10^{-5}$) and particulate domoic acid ($r = +0.33$, $p = 1\times10^{-3}$) at the NEMO mooring, and recovers documented PNW climate events (2014–16 marine heatwave, 2019 bloom) without supervision. Code and trained checkpoints are available at https://github.com/ansoncchen/DATect-Forecasting-Domoic-Acid under ocean anomaly detection/.

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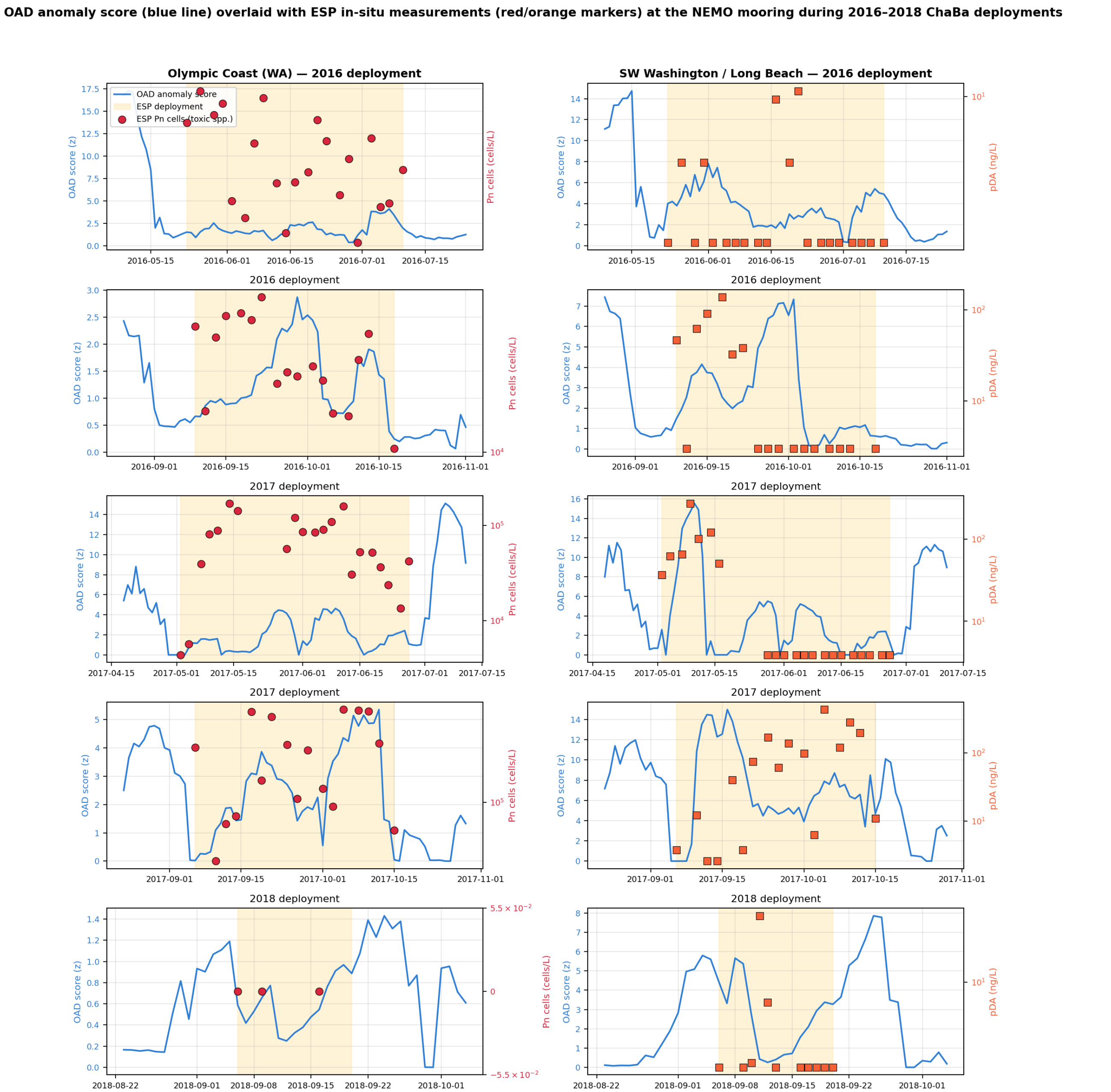

Figure 5. Time-series overlay of OAD score (blue line, left axis) and ESP in-situ measurements (red/orange markers, right axis on symlog scale) across the five 2016–2018 ChaBa deployment windows. Yellow shading marks each deployment (±14 days of context). Left column: Olympic Coast OAD vs. ESP Pn cells. Right column: SW Washington OAD vs. ESP pDA. The temporal co-evolution of the unsupervised satellite signal and independent in-situ biology is direct visual confirmation that the OAD representation captures real bloom dynamics at the offshore source.

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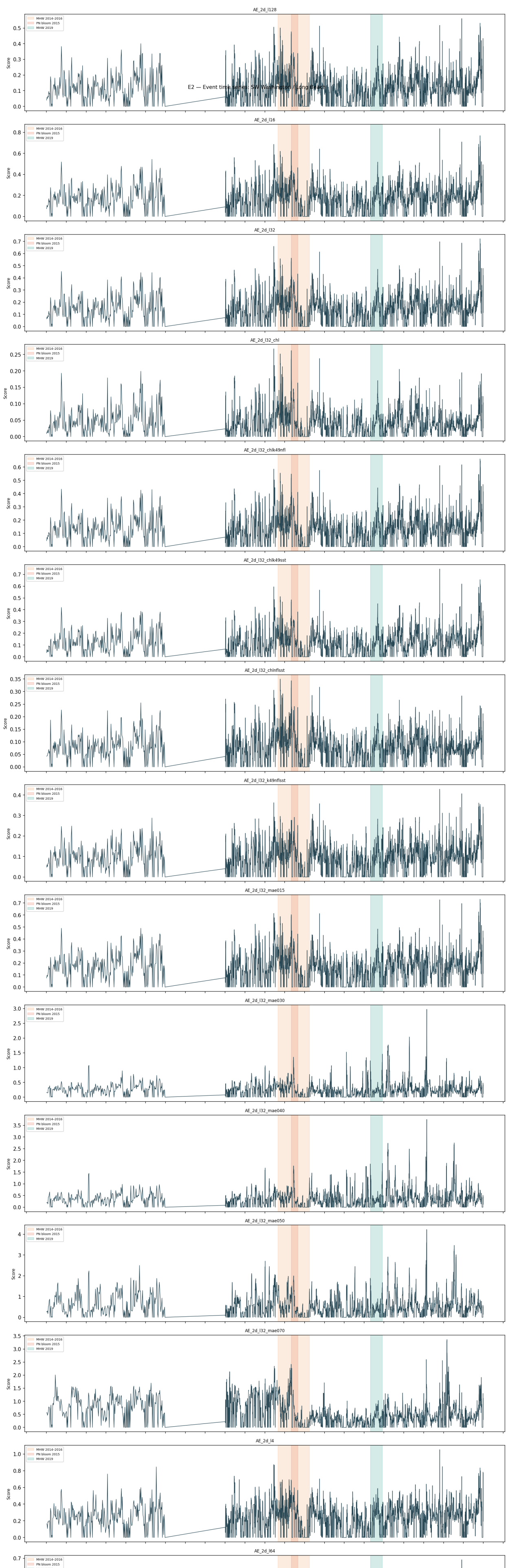
AE_2d_l128
MHW 2014–2016
PN bloom 2015
MHW 2019
Score
E2 — Event time series: SW Washington / Long Beach
AE_2d_l16
AE_2d_l32
AE_2d_l32_chl
AE_2d_l32_chlk49nfl
AE_2d_l32_chlk49sst
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